Test Edge Cases and Fairness

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

Recall and choose your starting point

> Record your starting explanation for Test Edge Cases and Fairness before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Test Normal Cases

Without opening the old lesson, explain one key idea from Test Normal Cases. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Design the Decision Process

Without opening the old lesson, explain one key idea from Design the Decision **Process**. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Humans Set Boundaries

Without opening the old lesson, explain one key idea from Humans Set Boundaries. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Test Edge Cases and Fairness trace and record the first step you can explain without help.

Use W2: Check the validation **boundary** for Testing boundaries and varied conditions. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Test Edge Cases and Fairness explanation against the supplied facts before discussing it.

Use the worked case for Testing boundaries and varied conditions. Proposed explanation: “At the limit means outside the rule”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Find an independent capacity boundary

A new equipment selector lists E1 first, capacity 4; E2 second, capacity 8. Valid group size is a whole number from 1 to 10.

Choose the first item whose capacity is at least the group size, or no match; **invalid input** needs clarification. A faulty version uses capacity strictly greater than group size.

Independently give correct and faulty **output**s for groups 3, 4 and 5, then for group 8. Explain the defect and give the correct response to group 0. No booking occurs.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Test Edge Cases and Fairness target to a trace and a justified conclusion.

Before testing, define outcomes for clean paper, metal foil, mixed packaging and an unreadable card. The team later changes mixed packaging's expected result to match its sorter. Explain the evaluation problem.

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Test Edge Cases and Fairness case and show the deciding evidence.

A sorter handles clean paper but fails on crumpled paper and mixed packaging.

Design revealing tests.

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

> A sorter works on clear photos. Test a blurred pencil, a dim book and an unfamiliar lunch box. What makes these useful tests?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 34 Test and reflect

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, **boundary** case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

WORKBOOK · 9 OF 21

Project checkpoint

> Justify what Test Edge Cases and Fairness adds to Helpful Classroom Sorter and keep the

> evidence.

Expand the sorter test suite with edge, missing-data, and fairness cases plus expected safe behavior.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

Before testing, define outcomes for clean paper, metal foil, mixed packaging and an unreadable card. The team later changes mixed packaging's expected result to match its sorter. Explain the evaluation problem.

> **My independently reasoned answer**

Record support used: none, hint or teacher explanation.

W1 Plan the equality tests

> Plan the equality tests. Apply the supplied Test Edge Cases and Fairness facts and keep your reasoning.

For catalogue B version 1, write expected **output**s and reasons for 9 Yes, 10 Yes and 11 Yes. Explain why keeping quiet and catalogue fixed helps examine the time **boundary**.

Predict what the faulty strictly-less-than S comparison does for 10 Yes.

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Check the validation boundary

Check the validation boundary. Apply the supplied Test Edge Cases and Fairness facts and keep your reasoning.

Classify these requests before searching: 0 Yes, 30 Yes, 31 Yes, blank minutes with Yes, 10.5 Yes, and 10 Maybe. Give each expected **output** or clarification and explain whether any candidate should be checked.

Why must zero not be treated like a blank?

My reasoning and evidence

W3 Run a labelled catalogue variation

> Run a labelled catalogue variation. Apply the supplied Test Edge Cases and Fairness facts and keep your reasoning.

Copy B version 1 and change only B1's duration to 20, labelling it B version 2. Write the expected result for 12 Yes, then run your saved P1 and the manual baseline on this copy. Record the actual traces separately.

Explain why an answer checked against B version 1 would not be a fair judgement of this run.

> **My reasoning and evidence**

W4 Find an independent capacity boundary

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Find an independent capacity boundary. Apply the supplied Test Edge Cases and Fairness facts and keep your reasoning.

A new equipment selector lists E1 first, capacity 4; E2 second, capacity 8. Valid group size is a whole number from 1 to 10.

Choose the first item whose capacity is at least the group size, or no match; **invalid input** needs clarification. A faulty version uses capacity strictly greater than group size.

Independently give correct and faulty **output**s for groups 3, 4 and 5, then for group 8. Explain the defect and give the correct response to group 0. No booking occurs.

My reasoning and evidence

WORKBOOK · 16 OF 21

W5 Audit what changed in the test

> Audit what changed in the test. Apply the supplied Test Edge Cases and Fairness facts and keep your reasoning.

A team runs 15 No on B version 3, gets B4, but records “B version 1” and calls B4 correct. Explain the documentation error and the expected result under each catalogue.

State what must stay the same in a fair prototype-versus-baseline comparison and why a test not yet run cannot be counted as a pass.

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

Pictorial case practice: Test Edge Cases and Fairness

> A fictional craft-book chooser accepts requests for whole available minutes from 0 to 20. Explain the labelled step before checking the changed condition.

A fictional craft-book chooser accepts requests for whole available minutes from 0 to 20.

Booklet A takes 6 minutes; B takes 12. Return first duration at most available time.

Check 5,6,7; also 0,20,21 and blank. Same catalogue applies to every user.

AI ACADEMY / AI-1/34

Test Edge Cases and Fairness

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Near a boundary

At 5: none. At 6 and 7: A.



2. Valid extremes

0: none. 20: A, because first-fit order remains fixed.



3. Invalid inputs

21 and blank require clarification, not ordinary search.



4. Coverage record

Record each user condition tested; do not invent unrun outcomes.

CHANGE ONE THING

Faulty code uses duration strictly below available time.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Which single **boundary** run most directly detects the strict-comparison bug, and why?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Add sorter **boundary** cases, missing fields and relevant user conditions to the test plan before making fairness claims.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Helpful Classroom Sorter record.

Project: Helpful Classroom Sorter

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

Add sorter **boundary** cases, missing fields and relevant user conditions to the test plan before making fairness claims.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional craft-book chooser accepts requests for whole available minutes from 0 to 20.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Record each user condition tested; do not invent unrun outcomes.

1. Near a **boundary**

2. **Invalid input**s

3. Coverage record

4. Valid extremes

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Coverage record Record each user condition tested; do not invent unrun outcomes. Adjacent values below, at and above 6 expose an accidental strict inequality.

Zero is valid even though no activity fits. The top valid **input** does not require selecting the longest activity: catalogue order still controls selection.

Coverage also asks whose conditions were tested, which cannot be inferred from these numbers alone.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Faulty code uses duration strictly below available time. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. We can decide the result without examining the changed **input** or the procedure.

3. Request 6 now wrongly returns none.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Request 6 now wrongly returns

none. A’s duration equals the available time and should be eligible under at most.

Repair a missing building block

Review Test Normal Cases (ai-1/33). Goal: Plan routine cases with expected results derived from the requirements. Short practice: Why is none correct for 8 but incorrect for 6? Use both capacity cards..

Return to this week and explain what you changed.

Review Design the Decision **Process** (ai-1/31). Goal: Design a decision **process** that covers valid, invalid and no-match requests.

Short practice: Where does the first original rejection occur, and why is K3 not originally selected?. Return to this week and explain what you changed.

Review Humans Set Boundaries (ai-1/28). Goal: Distinguish a system’s capability, the correctness of its **output** and permission to take an action. Short practice: Which original proposal is authorised?

Why do the other two need separate approval?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

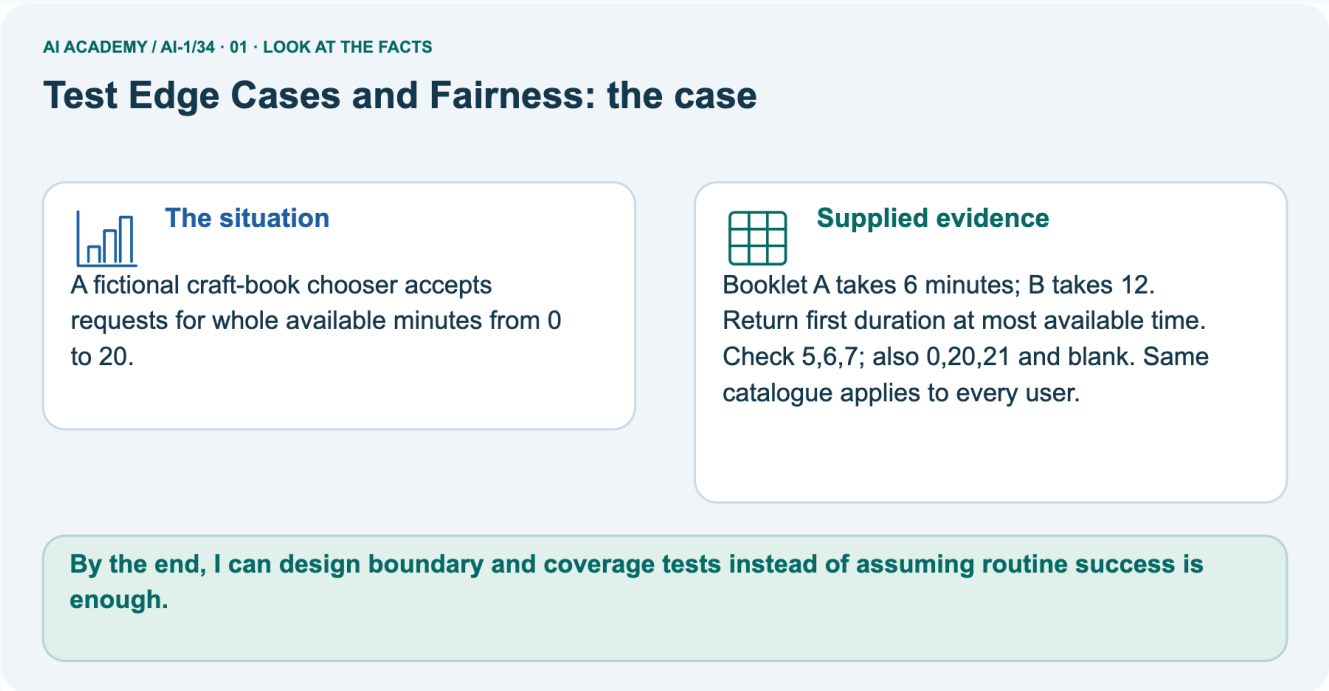
Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

Visual concept journey

> Explain each Test Edge Cases and Fairness picture from its labels, then record a relevant limit.

01. Test Edge Cases and Fairness: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

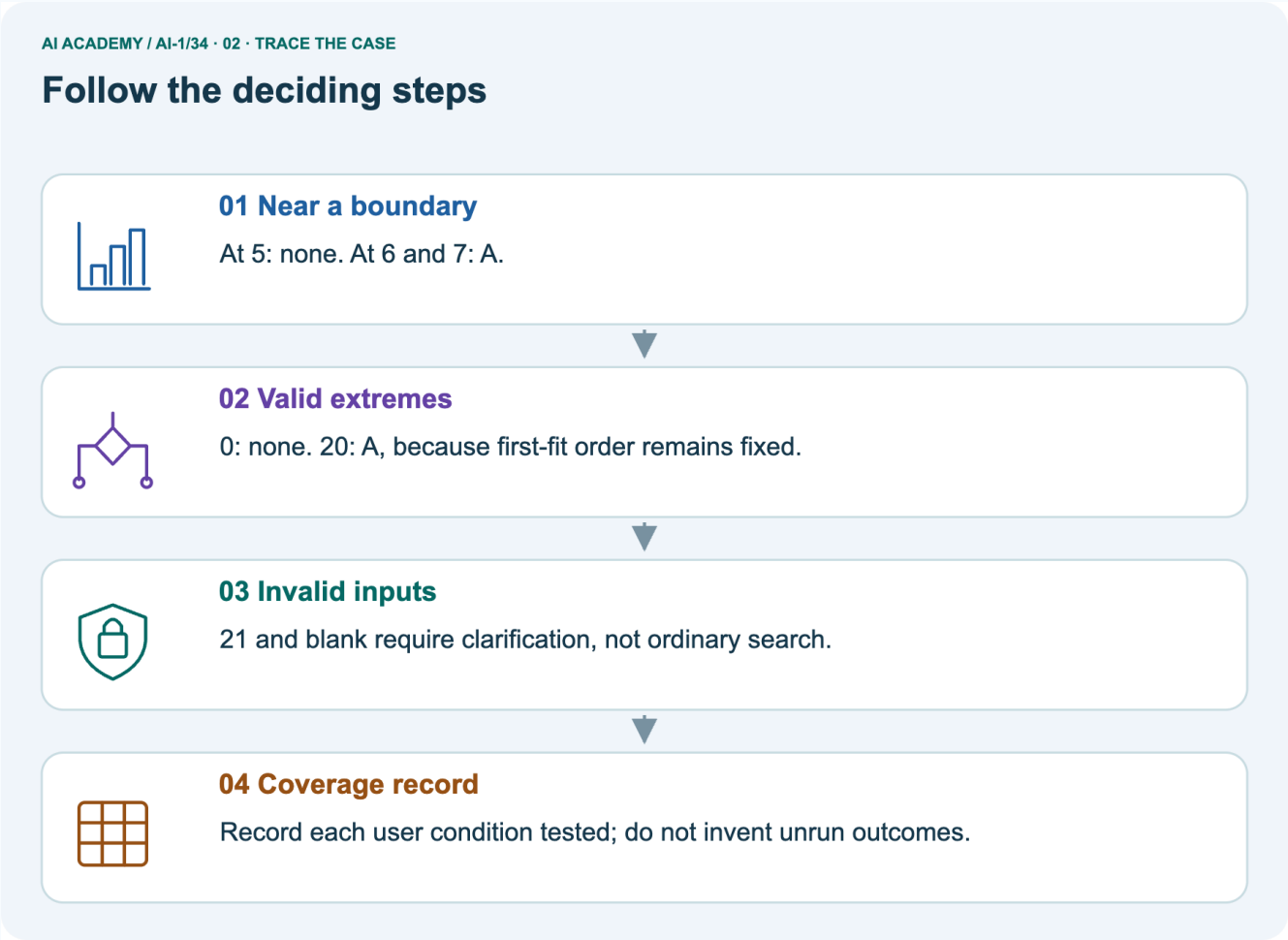
A fictional craft-book chooser accepts requests for whole available minutes from 0 to 20. Supplied facts: Booklet A takes 6 minutes; B takes 12. Return first duration at most available time.

Check 5,6,7; also 0,20,21 and blank. Same catalogue applies to every user.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

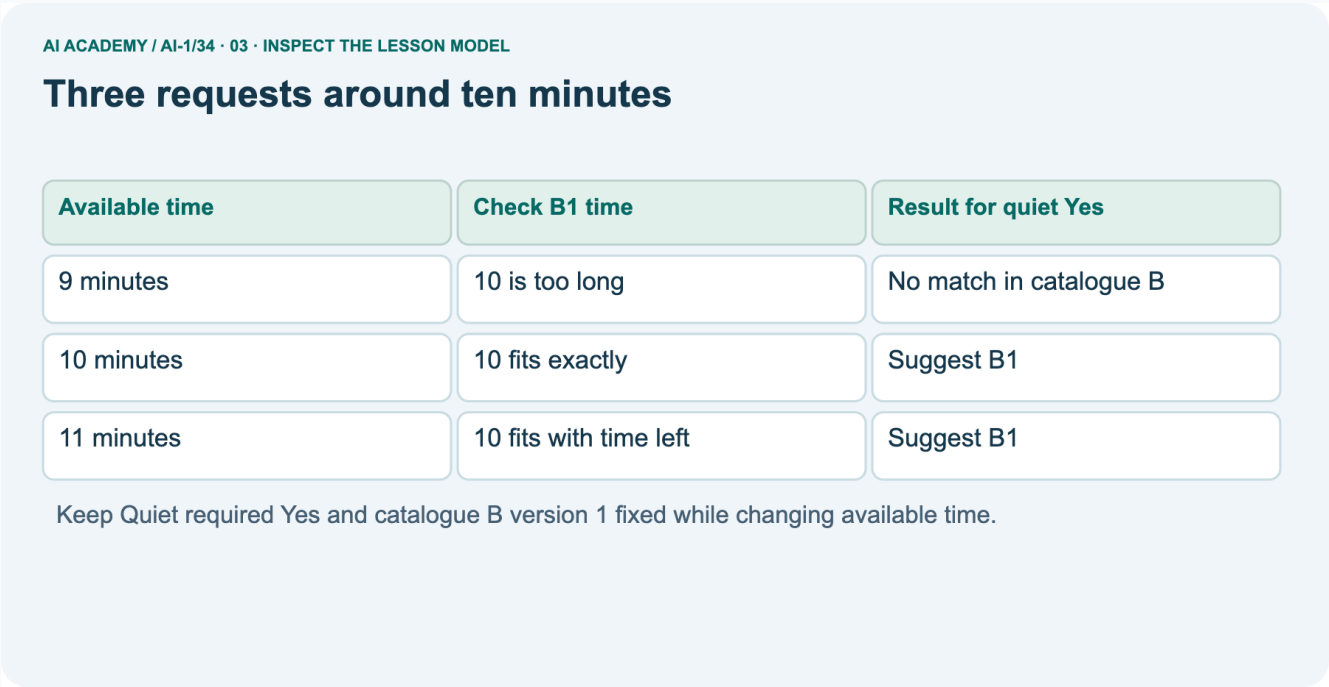
Near a **boundary**: At 5: none.

At 6 and 7: A. → Valid extremes: 0: none. 20: A, because first-fit order remains fixed. → **Invalid input**s: 21 and blank require clarification, not ordinary search. → Coverage record: Record each user condition tested; do not invent unrun outcomes.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Three requests around ten minutes



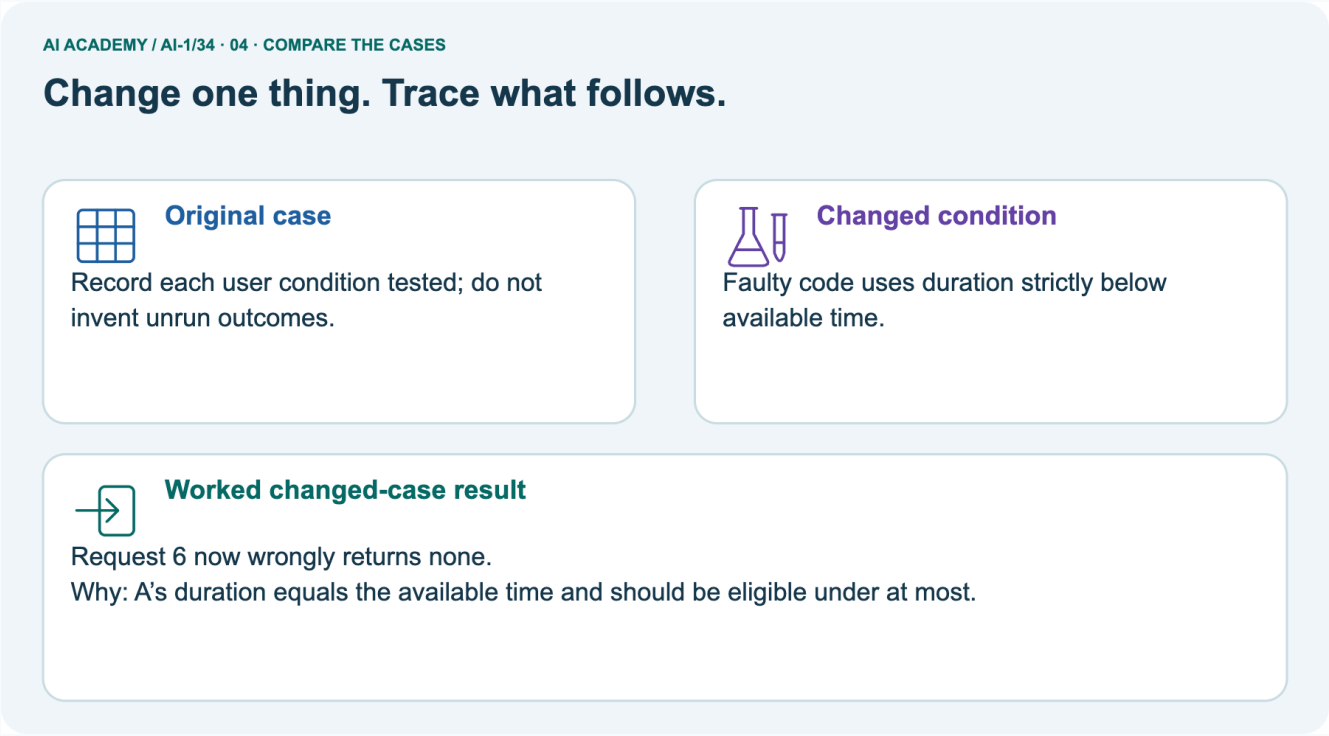
Follow the labelled picture with the matching explanation. Open the vector diagram

Three requests around ten minutes. 9 minutes | 10 is too long | No match in catalogue B. 10 minutes | 10 fits exactly | Suggest B1. 11 minutes | 10 fits with time left | Suggest B1

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

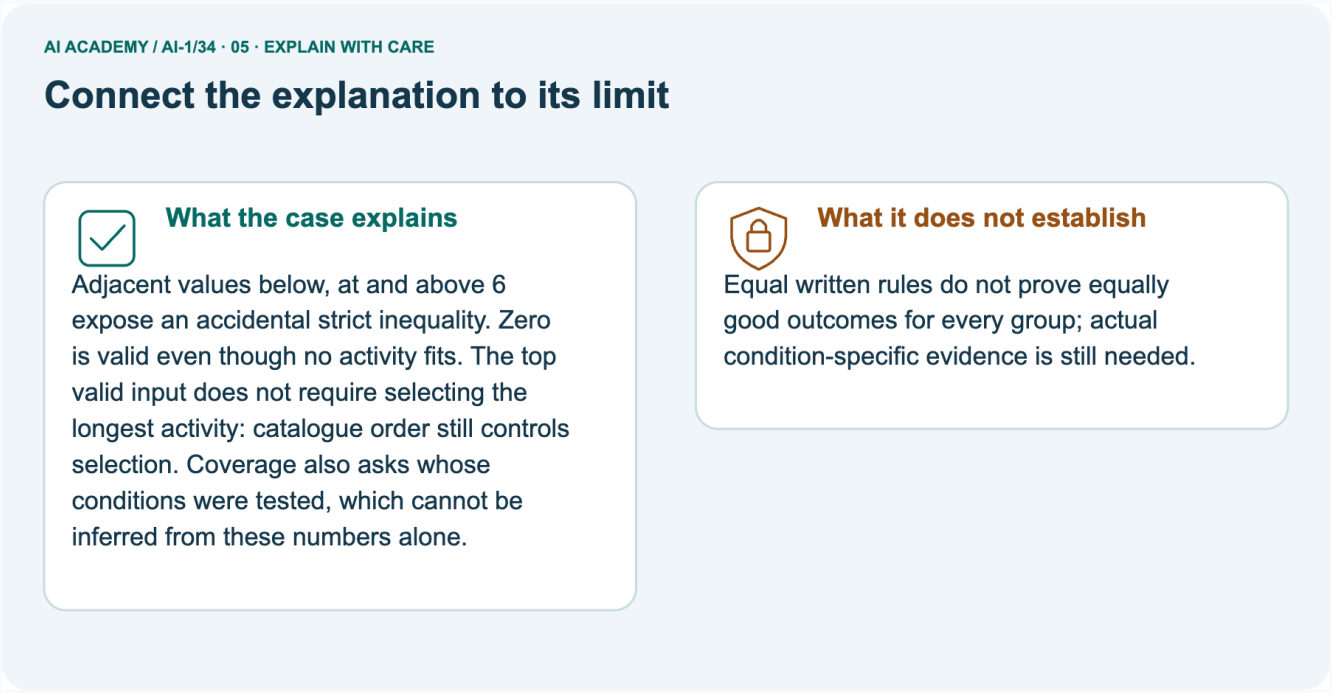
Original case: Record each user condition tested; do not invent unrun outcomes. Changed condition: Faulty code uses duration strictly below available time. Expected result: Request 6 now wrongly returns none.

Reason: A's duration equals the available time and should be eligible under at most.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Adjacent values below, at and above 6 expose an accidental strict inequality. Zero is valid even though no activity fits.

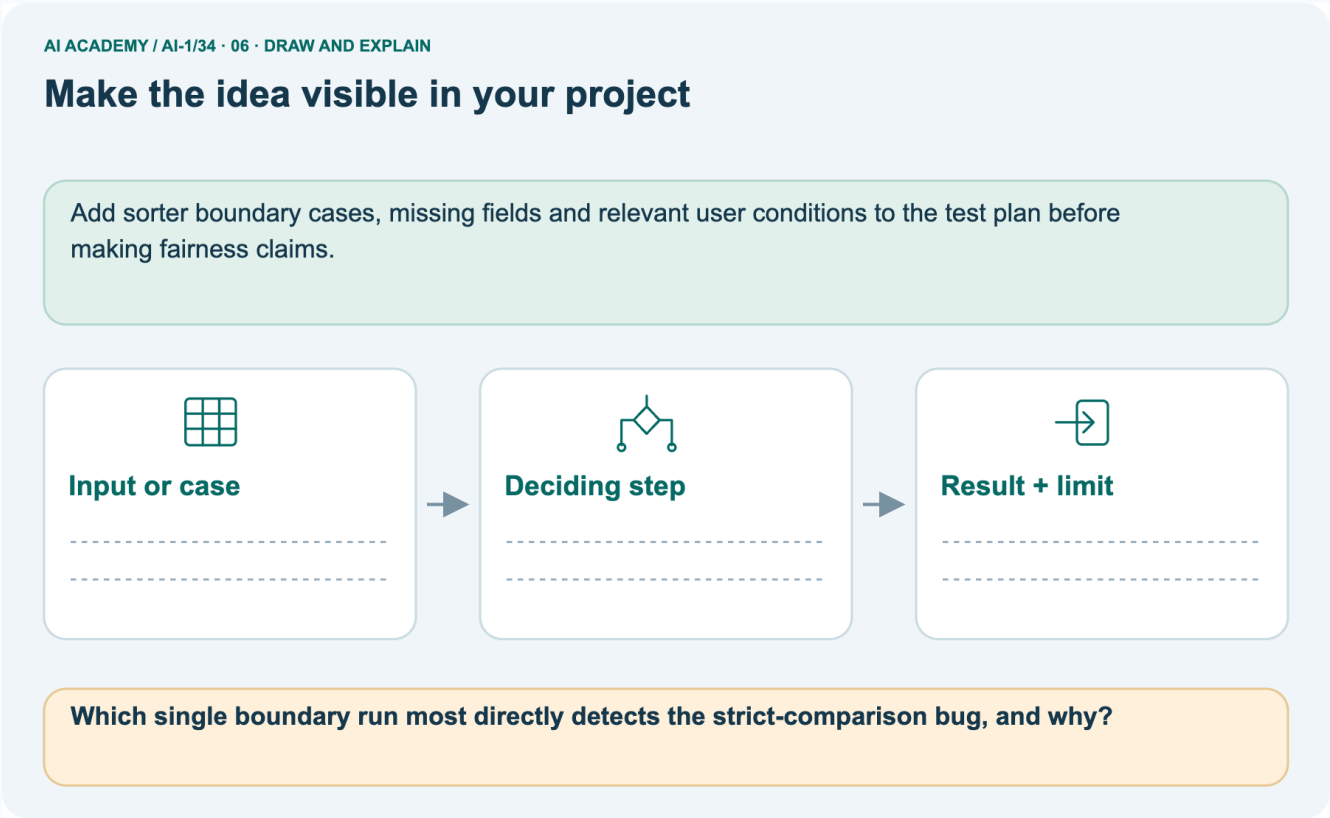
The top valid **input** does not require selecting the longest activity: catalogue order still controls selection. Coverage also asks whose conditions were tested, which cannot be inferred from these numbers alone.

Limit: Equal **written rule**s do not prove equally good outcomes for every group; actual condition-specific **evidence** is still needed.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Add sorter **boundary** cases, missing fields and relevant user conditions to the test plan before making fairness claims. Draw your **input**, deciding step, result and limit.

Supported question: Which single **boundary** run most directly detects the strict-comparison bug, and why?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.